The line with doubled origin is not separated

Rings are commutative. Schemes are represented, as in [1, 3], by locally ringed lattices: a distributive lattice L of (compact) opens with a sheaf of rings $\mathcal{O}$ on it. For a ring A, $\operatorname{Spec} A$ is the Zariski lattice $\operatorname{Zar}(A)$, generated by the basic opens $D(f)$, with the structure sheaf $\mathcal{O}_A$; we use only that $\mathcal{O}_A(1) = A$, that $\mathcal{O}_A(D(f)) = A[1/f]$, and that the restriction $\mathcal{O}_A(1) \to \mathcal{O}_A(D(f))$ is the canonical map $A \to A[1/f]$. A locally ringed lattice X is *affine* when the canonical map $X \to \operatorname{Spec}\Gamma(X)$, with $\Gamma(X) = \mathcal{O}(1)$, is an isomorphism, and an open u is *affine* when the restriction $X|_u$ is affine.

1 Separated schemes

For opens u, v of X write $r\colon \mathcal{O}(u) \to \mathcal{O}(u \wedge v)$ and $r'\colon \mathcal{O}(v) \to \mathcal{O}(u \wedge v)$ for the restriction maps.

Definition 1. *The overlap $u \wedge v$ is* generated *by u and v when every $z \in \mathcal{O}(u \wedge v)$ is a finite sum*

$$z = r(a_1)\, r'(b_1) + \cdots + r(a_n)\, r'(b_n), \qquad a_i \in \mathcal{O}(u),\ b_i \in \mathcal{O}(v),$$

that is, when $\mathcal{O}(u) \otimes \mathcal{O}(v) \to \mathcal{O}(u \wedge v)$ is surjective. X is separated *when for any two affine opens u and v the overlap $u \wedge v$ is affine and generated by u and v.*

Remark 1. Classically, a scheme is separated, i.e. its diagonal is a closed immersion, exactly when this holds for all affine opens u, v: the diagonal restricted to $\operatorname{Spec}\mathcal{O}(u) \times \operatorname{Spec}\mathcal{O}(v)$ is the closed immersion $\operatorname{Spec}\mathcal{O}(u \wedge v) \to \operatorname{Spec}(\mathcal{O}(u) \otimes \mathcal{O}(v))$. The definition above avoids covers, products of schemes and tensor products.

2 The line with doubled origin

Let k be a ring, $A = k[X]$ and $d = D(X) \in \operatorname{Zar}(A)$, the punctured line. The line with doubled origin glues two copies of $\operatorname{Spec} A$ along d by the identity. Its lattice is

$$L = \{(u_0, u_1) \in \operatorname{Zar}(A) \times \operatorname{Zar}(A) \mid u_0 \wedge d = u_1 \wedge d\},$$

with componentwise order and operations, and its sheaf is

$$\mathcal{O}(u_0, u_1) = \mathcal{O}_A(u_0) \times_{\mathcal{O}_A(m)} \mathcal{O}_A(u_1), \qquad m = u_0 \wedge d = u_1 \wedge d,$$

the pairs (s, t) with $s|_m = t|_m$, restricted componentwise. The two charts are

$$e_0 = (1, d), \qquad e_1 = (d, 1),$$

the two copies of the line; each is isomorphic to $\operatorname{Spec} A$, hence affine. Their overlap is $e_0 \wedge e_1 = (d, d)$, and $\mathcal{O}(e_0 \wedge e_1)$ consists of the pairs (s, t) of elements of $\mathcal{O}_A(d) = A[1/X]$ with $s = t$.

Lemma 1. *Let $\rho\colon A \to C$ and $\pi\colon B \to C$ be ring maps. The elements $z \in B$ such that $\pi(z)$ is in the image of ρ form a subring of B. In particular they are closed under finite sums of products.*

Proof. The image of a ring map is a subring, and π^{-1} of a subring is a subring. □

Theorem 1. *If the line with doubled origin over k is separated, then $1 = 0$ in k.*

Proof. Assume it is separated. The charts e_0 and e_1 are affine, so their overlap (d, d) is generated by them. Let $\rho\colon A \to A[1/X]$ be the canonical map and $\pi\colon \mathcal{O}(d,d) \to A[1/X]$ the first projection.

Sections of e_0. An element of $\mathcal{O}(e_0)$ is a pair (a_0, a_1) with $a_0 \in \mathcal{O}_A(1) = A$. The first component of its restriction to (d, d) is $a_0|_d = \rho(a_0)$.

Sections of e_1. An element of $\mathcal{O}(e_1)$ is a pair (b_0, b_1) with $b_0 \in \mathcal{O}_A(d)$, $b_1 \in \mathcal{O}_A(1) = A$, and $b_0|_m = b_1|_m$ for $m = d \wedge d = d$. So $b_0 = b_1|_d$, and the first component of the restriction of (b_0, b_1) to (d, d) is $\rho(b_1)$.

Hence π maps the restrictions of both $\mathcal{O}(e_0)$ and $\mathcal{O}(e_1)$ into the image of ρ, and by Lemma 1 so does it map every finite sum of products $r(a_1)r'(b_1) + \cdots + r(a_n)r'(b_n)$.

Now take $z_0 = (1/X, 1/X) \in \mathcal{O}(d, d)$. By generation, $1/X = \rho(p)$ for some polynomial $p \in k[X]$. Then $\rho(Xp) = X \cdot \frac{1}{X} = 1 = \rho(1)$. The map ρ is injective, since X is a regular element of $k[X]$: if $q/1 = 0$ in $A[1/X]$ then $X^n q = 0$ for some n, so $q = 0$. Hence $Xp = 1$ in $k[X]$, and comparing constant coefficients gives $0 = 1$ in k. □

Only the generation condition is used, and only at one pair of affine opens. The overlap (d, d) is in fact affine, isomorphic to $\operatorname{Spec} k[X, X^{-1}]$, so it is the second condition that fails: the two copies of $k[X]$ both map onto the same subring $k[X]$ of $k[X, X^{-1}]$, which does not contain $1/X$. Generation is a proposition (the sums are merely given), and the conclusion $1 = 0$ is a proposition since k is a set, so the argument is constructive as written. In particular, over a ring with $1 \neq 0$ the doubled line is not separated.

3 What has been formalised

The definition and the theorem are checked in the proof checker Unit [2], in the development `sag/qcqs` of qcqs schemes as locally ringed lattices.

- `SchemeSeparated.seg` defines `Overlap[u,v].Generated` (every section of the overlap is merely a finite sum of products of restrictions) and `IsSeparated`, with "affine open" the existing notion `Qcqs.AffineOpen`.
- `DoubledLineNotSeparated.seg` proves

  `separated-one-zero` : `IsSeparated` $\to 1 = 0$

  for the doubled line `Dbl.doubled` of `DoubledLine.seg`, over any ring k. It follows the proof above: the lifting lemma, the two chart computations (the second uses the agreement of the components on the middle open), the element z_0 built from the inverse of X in the localisation, and injectivity of $k[X] \to k[X, 1/X]$ with evaluation at 0 (`VariableKernel.seg`). The affineness of the charts is `DoubledLineQcqs.seg`.
- Not formalised: that the overlap (d, d) is affine (not needed), and the classical equivalence of the definition with the diagonal being a closed immersion (the definition is taken as primitive).

The development `sag/qcqs` checks with 16 141 declarations and no failure, in about 19 s. No new axiom is used; the parameters of the development are univalence, images and surjections, as for the rest of `sag/qcqs`. The doubled line is also checked to be a qcqs scheme (`DoubledLineQcqs.seg`) and not affine over a nonzero ring (`DoubledLineNotAffine.seg`).

References

[1] T. Coquand, H. Lombardi and P. Schuster. Spectral schemes as ringed lattices. *Annals of Mathematics and Artificial Intelligence* 56 (2009), 339–360.

[2] T. Coquand. UNIT: a type checker for dependent type theory. `https://github.com/coquand/unit`.

[3] M. Zeuner. Univalent foundations of constructive algebraic geometry. Stockholm University, 2024.